

EDEM Tutorial

Rock Box

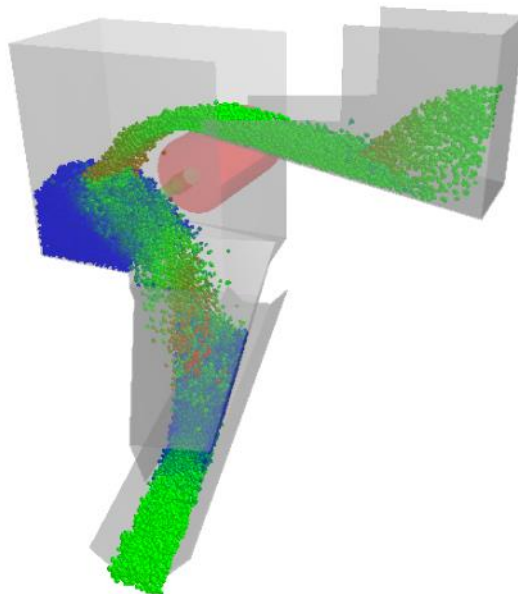
Introduction

This introductory tutorial shows how to set-up and analyze an EDEM simulation. The model used here is a rock box conveyor transfer system.

The tutorial consists of two parts – the first part (**page 3**) describes a simulation setup suitable for using the Altair Student Edition. It can be established quickly and the simulation time is low (~2 minutes on 2 CPU cores). Following this the simulation is modified to be a high fidelity model is described in part 2 (**page 16**). This extended simulation requires a CUDA compatible GPU to run and isn't suitable for the Student Edition.

The focus of this tutorial is on:

- Using the GEMM database
- Creating multiple particle types and shapes
- Using mixed shapes (Polyhedral particles and Multi-Sphere)
- Importing a particle size distribution from a *.csv file
- Implementing conveyor kinematics to the conveyor geometries
- Analyzing the geometry to choose appropriate parameters for velocities of conveyor belts
- Creating a mass flow sensor and a bin group
- Generating graphs and pie charts to analyze simulation data



PART I – Initial setup

1. Start EDEM.
2. Go to File > Save As...
3. Select a location on the local machine (e.g. **C:\EDEM_Tutorials**).
4. Enter a file name (e.g. **Rock_box.dem**) and click Save.

Please note this tutorial has associated CAD files rock_box1.stl, Rock_Template1.stl, Rock_Template2.stl and Rock_Template3.stl. In addition, a particle size distribution file is available (PSD.csv) It is recommended that you copy these files to the same location as the simulation files.

EDEM Creator: Setting up the model

Step 1: Set the Global Model Parameters

Choose the units

The first step in creating the model is to set the units used throughout EDEM.

1. Go to the Tools > Options... menu and select the Units tab.
2. Change the following measurement units:
 - Angle to degrees(**deg**)
 - Angular velocity to rad/s
 - Length to **mm**
 - Mass to **kg**
 - Mass Flow Rate to **kg/s**
 - Velocity to **m/s**
3. Click OK.

Enter the model title and description

1. Click on Project in the Creator Tree.
2. Enter the title **Rock Box** in the Title box under the Simulation section.
3. Enter a description in the Description field.

Step 2: Define the Bulk and Equipment Materials

Adding materials

The first step is to add the materials which will be used in the simulation.

1. Right click on the Bulk Material in the Creator Tree and select Open GEMM Wizard. This will open the Generic EDEM Material Model (GEMM) Database.
2. Select Search by Material Properties and click Next.

The GEMM wizard will aim to select a material appropriate for the simulation based on the three inputs.

1. Select the **Extra Large** option for the scale of the application. Click Next.
2. Select the **1000 - 1500 kg/m³** option for the bulk density. Click Next.
3. Enter an angle of repose of **38.00°**. Click Next.

The GEMM wizard will present the 10 closest matches from the GEMM database, based on the three options previously entered.

4. Select the top entry, with material properties as shown below

Generic EDEM Material Model (GEMM) Retrieval

JKR (J/m ²)	P to P Coefficient of Restitution	P to P Coefficient of Rolling Friction	P to P Coefficient of Static Friction	Angle of Repose (degrees)
0	0.15	0.05	0.44	38

If none of the 10 materials shown displays the values seen above, choose the material which is the closest match.

5. Click Next.
6. Select the **High Wall Friction** and **Medium Wall Friction** checkboxes in the Inclination Test Details section.
7. Click Finish.

The GEMM wizard will have now imported a Bulk Material and two Equipment Materials.

Expand the Bulk Material section in the creator tree to display the particle material name (e.g. GEMM_particle_material_11921).

8. Right click on the Bulk Material, select Rename Material and type **rock** in the New Name box.
9. Expand the Equipment Material section and repeat step 8 for the two Equipment Materials, renaming the medium and high friction materials **steel** and **rubber**, respectively.

The Bulk Material properties (interactions, shape and size distributions) have been set by the GEMM Database. Also interactions between the Bulk Material (rock) and Equipment Materials (rubber and steel) have already been created by the GEMM wizard. To check this, select **rock** in the Creator Tree and click on the pull-down menu in the interactions section to see the interactions that have been defined.

Step 3: Define the Geometry

The next step is to define the rock box geometry used in the model.

Import the rock box geometry

The rock box geometry has already been created in a CAD package and is ready to be imported into EDEM.

1. Right click on the Geometries section in the Creator Tree.
2. Click on Import Geometry.
3. Navigate to the file **rock_box1.stl** and import it.
4. When prompted, set the Units of measurement to **Millimeters**. Leave the Merge Sections option **unchecked**. Leave all the settings as the default values and click OK.

The geometry will appear in the viewer window. It is made up of 4 sections, which are listed in the Sections drop-down list. When any part is selected it is highlighted in red.

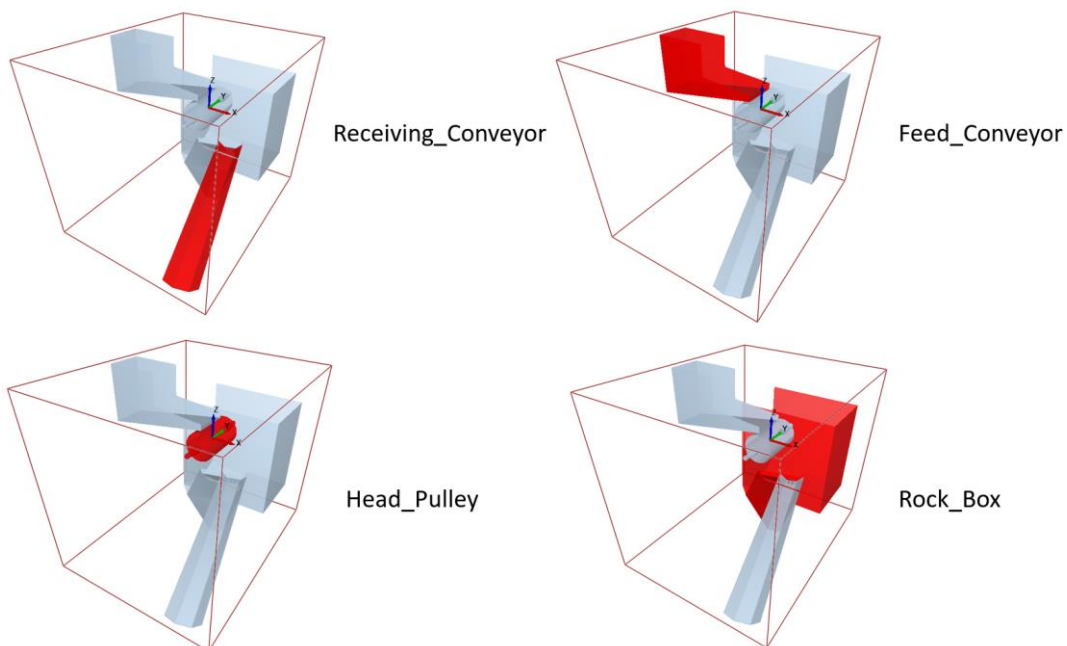
5. Re-name the geometry sections appropriately to:

Receiving_Conveyor

Head_Pulley

Feed_Conveyor

Rock_Box



6. Select the Rock_Box in the Geometries section of the Creator Tree.
7. Change the Material to **steel**, using the Material pull-down menu.

- Repeat steps 6 and 7 for the Head_Pulley and Conveyors, ensuring the material is set to **rubber**.

Setup the Head Pulley Motion

- Right click on the **Head_Pulley** in the Creator Tree.
- Hover over Add Motion and then select Add Linear Rotation Kinematic.
- Leave Start and End Time as default; modify the Initial Velocity to **7.08 rad/s** to match the linear velocity of the conveyor (3.54 m/s) since the radius of the pulley is 0.5m.
- To specify the Axis of Rotation, enter the start and end points as shown below:

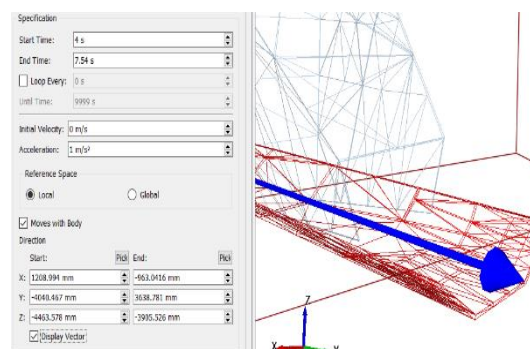
Start:	Pick	End:	Pick
X: 0 mm		0 mm	
Y: -1585 mm		1585 mm	
Z: 0 mm		0 mm	

- Check the **Display Vector** checkbox to display the axis of rotation.
- The axis could also have been specified by clicking the **Pick** buttons next to Start and End. We will use this approach later when specifying conditions for the conveyors. We enter the points manually here to make sure that the axis is defined accurately, otherwise the axis of rotation would move and the simulation would not produce the desired qualitative results.

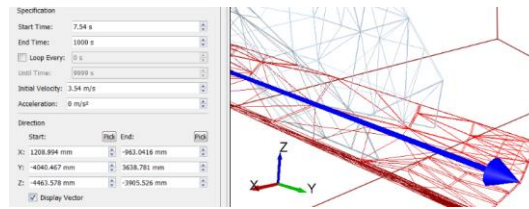
Configure the moving conveyors

The motion of the conveyors is taken into account using the Conveyor Translation Kinematic. The kinematic adds a linear velocity to the geometry section.

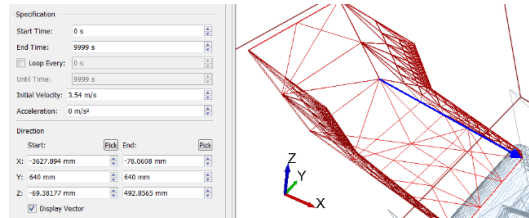
- Right Click on **Receiving_Conveyor** and select Add Motion > Add Conveyor Translation
- Set the initial acceleration as 1 m/s^2 starting at 4 s and ending at 7.54 s, and then using the pick tool set the direction as follows. Hint – when using the pick tool set the geometry to display mesh and pick inside the triangle close to the node you want to select.



- Right click on the kinematic to 'copy motion' keeping the default options. Edit to create a second conveyor kinematic for the Receiving Conveyor with a linear velocity 3.54 m/s starting at 7.54 s as shown:



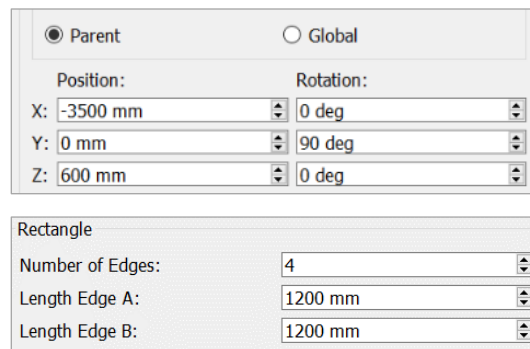
4. Select the **Feed_Conveyor** and setup a conveyor kinematic with initial linear velocity of 3.54 m/s as shown:



Create the particle factory plate

Particle factories are used to define where, when and how particles appear in a simulation. All factories must be based on a section of geometry (whether 'real' or 'virtual'). This defines the area in the model that produces the particles.

1. Right click on the Geometries section in the Creator Tree > Add Geometry > **Polygon** and then enter the name **factory_plate** into the highlighted field.
2. Select the factory_plate and in the General section set the type to **Virtual**, since the plate is not a physical part of the machinery.
3. Expand the factory_plate section and select the Transform and Polygon subsections. Set the Number of Edges, X/Y/Z Center and Dimensions of the polygon as follows:



Create the particle factory

1. Right click on the factory_plate section in the Creator Tree.
2. Click Add Factory. Select **New Factory 1** and make sure the Factory Type is set to **dynamic**. If it is not, right click on the new factory in the Creator Tree and click Change Factory Type.
3. In the Particle Generation section, Select **Unlimited Number** of particles.
4. In the Generation Rate section, select **Target Mass** and set the rate to **220kg/s**.

The Material will automatically default to the only Bulk Material created in this simulation which is rock.

5. Check that the Velocity is set to **fixed** then click the configure button  to set the velocity in the X-direction to **3.54 m/s**. The particles will leave the factory with this velocity.

EDEM Simulator: Running Simulation

Once the model has been set-up in the Creator we move to the Simulator to run it.

1. Click on the Simulator button  on the toolbar.

Step 1: Set the Time Options

Set the time step

The time step is the amount of time between iterations (calculations) in the Simulator.

The particles in this model are loosely packed for the majority of the simulation, so set a time step of 20% of the Rayleigh Time Step.

1. If checked, **uncheck** the Auto Time Step checkbox.
2. Set the Fixed Time Step upper box to **20%**. The numerical value of the time step would be calculated automatically.

Set the simulation time and data write-out frequency

The simulation time is the amount of real time your simulation represents.

1. In the Simulation Time section, set the Total Time to **15s**.
2. Set the Target Save Interval to **0.02s** to specify write-out frequency.

Usually it is not necessary to write a data point for every iteration in a simulation. Doing so will usually result in a slow simulation and a very large amount of data being collected.

Step 2: Set the Grid options

The grid options are used to optimize the simulation.

1. Set the Cell Size to **3 Rmin**. (Rmin = smallest particle radius in simulation)

Note the number of grid cells (Approx. Number of Cells) is stated below. The simulation results are not affected by the number of grid cells, only by the time taken to reach them. The more grid cells a simulation uses the more system RAM is required.

It is necessary to find a balance between RAM usage and simulation performance. For simulations with a narrow size distribution the ideal value will be between 2 - 3 Rmin, but for wider size distributions this will increase.

Step 3: Run the Simulation

You are now ready to start the simulation.

1. Click the start Progress button  at the bottom of the simulation window.
2. In Viewer Controls, click the **Refresh Viewer** button  or enable **Auto Update**  at any point to update the Viewer and see how the simulation is progressing.

EDEM Analyst: Analyzing Your Results

The Analyst is used to review, examine and analyze the results of the simulation.

1. Click on the Analyst button  on the toolbar.
2. Use the controls below the main window to rewind the simulation to the first time step (use the  button).
3. Click  button to play through each time step of your simulation.

Step 1: Configuring the Display

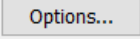
The options in the Display section of the Analyst Tree are used to configure how the different elements in your model appear.

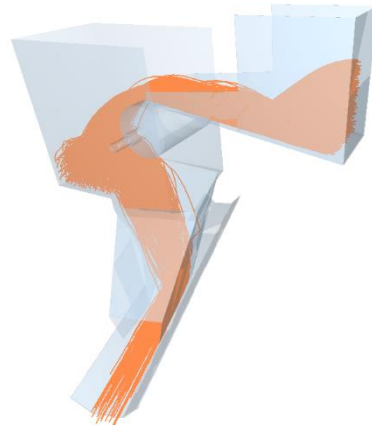
Configure the geometry

1. Expand the Display section in the Analyst Tree and then expand the Geometries subsection.
2. Experiment with altering the geometry display. To do this, in the Geometry section modify the Display Mode options. Display Mode can be chosen to be **Filled**, **Mesh** or **Points**. The Opacity can take values between **0-1**, with 0 being completely transparent and 1 being solid.
3. Click the Apply All button to see the changes in the Viewer window.
4. You can set the Display Mode to **Filled** with **0.3** Opacity to view the particles' flow. Click Apply All.
5. Before finishing, click on the factory_plate in the Geometries section and uncheck the **Display factory_plate** box. Click Apply if Auto Update is disabled, otherwise the change will be applied automatically.

Configure the particles

The display of the particles can be altered in a similar way to that of the geometry. In addition, the particles can be represented using vectors, cones, streams, etc.

1. Expand the Particles section in the Analyst Tree.
2. Select **Vector** from the Representation pull-down menu. Click Apply All.
3. Change time-steps to review how the material flows.
4. Change the Representation to **Stream**. Click the  button and in the Streams section, towards the bottom, check **Stream All Steps** and **Stream Fade** checkboxes.
5. Click OK and then the Apply All button.
6. Rewind  (if necessary) and Play  through the simulation.
7. Set the Representation back to **Default** before proceeding. Click Apply All.



Step 2: Coloring Elements

The particles, geometry and contacts within your model can be colored in a variety of ways. Any section of geometry, particle, contact type or selection group can be colored independently.

Color the geometry according to section type

1. In the Geometries section, select the **Feed_Conveyor**.
2. In the Coloring section, set the Color by to **Velocity**.
3. Set the Min, Mid and Max colors to the **Blue, Green** and **Red** respectively.
4. For Min Value and Max Value check the **Auto Update** boxes.
5. Repeat instructions 1 to 4 selecting **Receiving_Conveyor** and **Head_Pulley**.
6. Rewind the simulation, if necessary, and click the Play button to view the velocity of moving parts of the geometry.
7. Before continuing, select the **Geometries** section in the Analyst Tree.
8. In the Coloring section, set the Color by to **Uniform** and the Color to **Geometry Default**. Click Apply All.

Color the particles according to their velocity

1. Click on Particles in the Analyst Tree.
2. In the Coloring section, set the Color by to **Velocity**.
3. Set the Min, Mid and Max colors to the **Blue, Green** and **Red** respectively.
4. Click the  buttons next to the Min and Max Value fields – this reads the current values from the point in the simulation being displayed in the Viewer at the moment. Check the **Auto Update** option so that the values continue to update whilst the simulation is being played.
5. Click the **Apply All** button to color the particles.
6. Select the **rock** particle in the Analyst Tree.

7. Click the **Show Legend** checkbox.
8. Click the Apply button, if present.
9. Rewind the simulation back to the start.
10. Click the Play button and watch how the colors update as the simulation is played back.

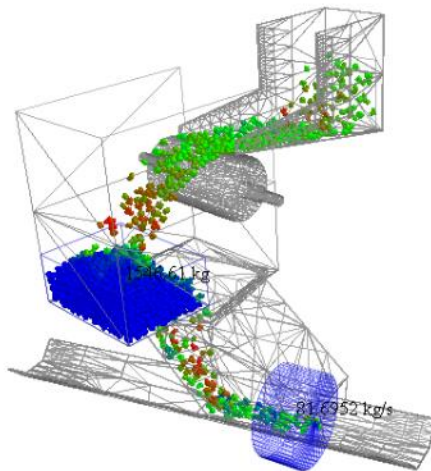
Step 3: Creating Selections

Selections allow data to be extracted from a particular area or element in the domain. Using selections, you can monitor any element passing through a particular area (known as a “bin”) or track particular elements wherever they move within the domain. You can then display, color, graph or export data based on the selections.

Create a mass flow sensor

We can examine the mass flow over time on the bottom conveyor by including a mass flow sensor.

1. Right click on Setup Selections in the Analyst Tree.
2. Hover over **Add Sensor Selection > Mass Flow Sensor**.
3. In the Representation section set Display Mode to **Always**.
4. Set the properties of the bin group as follows then click Apply:

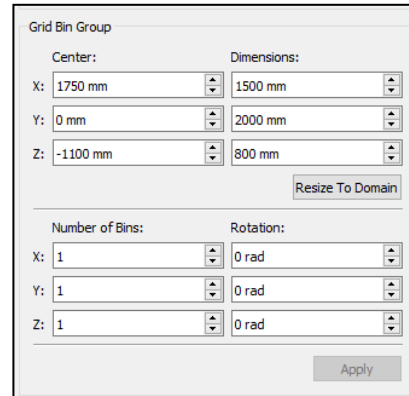


Geometry Bin	
Radius:	800 mm
Start Point: End Point:	
X: -20 mm	-200 mm
Y: 1300 mm	2000 mm
Z: -3800 mm	-3800 mm
Apply	

Create a grid bin group

We can also examine the increase of mass over time on the rock box. To do that, we need to create a bin group over it.

1. Right click on Setup Selections in the Analyst Tree.
2. Hover over Add Selection > **Grid Bin Group** and enter the name **Rock_Box Group** into the highlighted field. Click Ok.
3. Set the properties of the bin group as follows then click Apply:



The 'Grid Bin Group' dialog box is shown with the following settings:

Center:		Dimensions:	
X:	1750 mm	1500 mm	
Y:	0 mm	2000 mm	
Z:	-1100 mm	800 mm	

Buttons: **Resize To Domain**

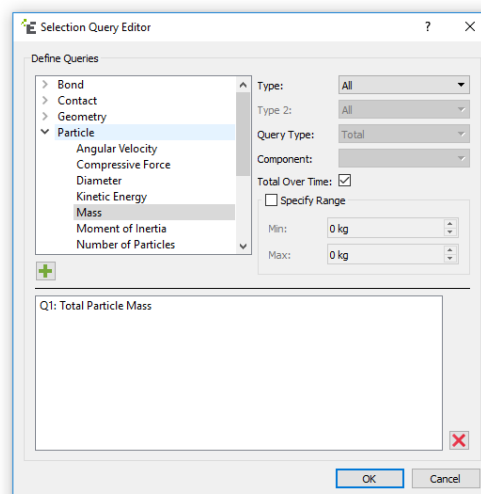
Number of Bins:		Rotation:	
X:	1	0 rad	
Y:	1	0 rad	
Z:	1	0 rad	

Button: **Apply**

Define selection group queries

A query is used to define a single element attribute, for example particle velocity, total force on a geometry section or number of collisions. To determine the total mass of particles accumulating in the rock box:

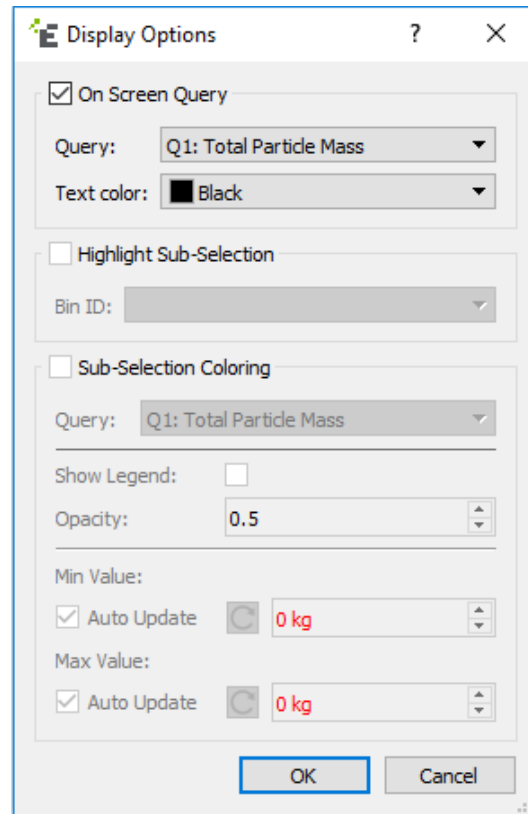
1. In View Controls, on the main toolbar, set the Current Time to 0s (the beginning of the simulation, when no particles are present in the rock box).
2. Select **Rock_Box Group** in the Setup Selections section.
3. Click the Queries Edit button in the Options section:
4. Expand the Particle tree then select **Mass**.
5. Select **All** in the Type menu. We could also select one of the particles in the Type Menu to only monitor that one particle type.
6. Set Query Type to **Total**.
7. Select the **Total Over Time** option then click the  button. The query is displayed in the list at the bottom of the dialog. Click OK.



8. Note that you do not need to create a query for the mass flow sensor.

Set selection group display options

1. Select **Rock_Box Group**.
2. Click the Display Options Edit button.
3. Tick the On Screen Query checkbox to display the mass in the viewer:



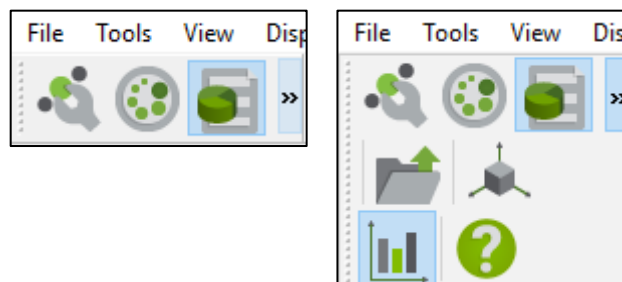
4. Click the play button to view the results in real time. As playback continues through the time-steps, bin group data is collected and displayed.

Step 4: Plotting the graphs

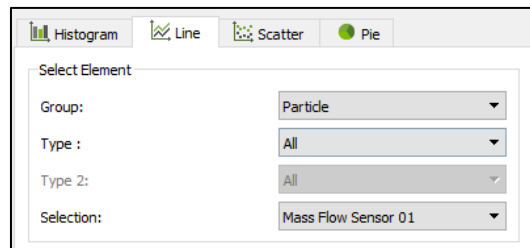
Plot a graph of mass flow rate on the bottom conveyor

To plot a linear graph of mass flow rate of all particles changing over time on the bottom conveyor:

1. Click the Create Graph button  in the toolbar. If you cannot see this on your toolbar, expand the Standard toolbar:



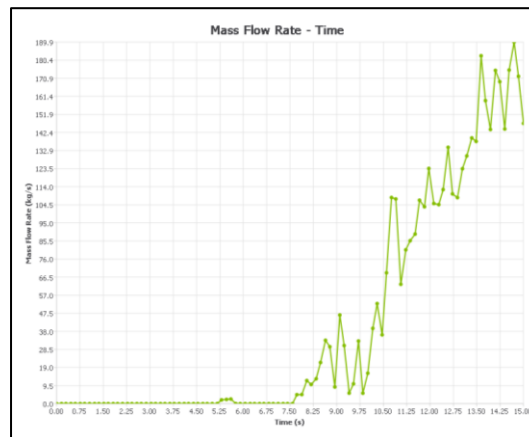
- Click on the Line Graph tab, set the parameters as follows:



Line Graph configuration window showing the following settings:

- Group: Particle
- Type: All
- Type 2: All
- Selection: Mass Flow Sensor 01

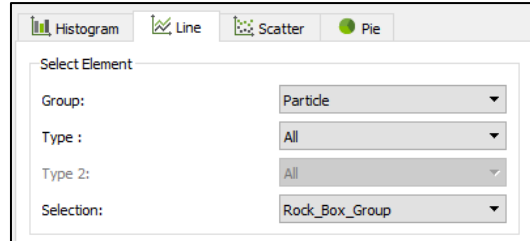
- In the X-axis tab, ensure that Time Range covers whole simulation time.
- Click the Y-axis tab. Set the Primary Attribute to **Mass Flow Rate**.
- Click Create Graph at the bottom to plot the graph.



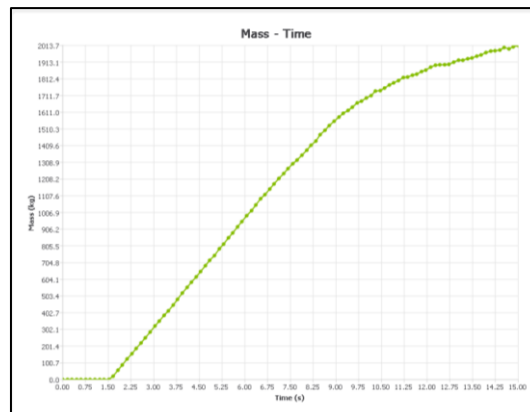
Plot the graph of total particle mass in the rock box

The bin group over the rock box registers the change of total mass of the particles there. To plot the increase in mass over time in the rock box:

1. In Line Graph tab, set the parameters as follows:



2. In the X-axis tab, ensure that Time Range covers whole simulation time.
3. Click the Y-axis tab. Set the Primary Attribute to **Mass** and the Component to **Total**.
4. Click Create Graph at the bottom to plot the graph.



From the graph, you can conclude that a steady state has not been reached yet.

PART II – Mixed Shape Solver

1. Open **Rock_box.dem** from Part 1.
2. Go to the EDEM Analyst, set Current Time to **0 s**.
3. Choose File > Export > Simulation Deck.
4. Choose a location (e.g. **C:\EDEM_Tutorials**) and file name (e.g. **Rock_box_full.dem**) and click Save.
5. In the Export Options dialog that appears, leave the default setting.

Note, this will create the simulation file at $t = 0$ s. An input deck can be taken from any point of a simulation. It is useful to sometimes take the input from the last timestep to store the simulation results or share with other users. This is a single timestep and much smaller in file size than the full simulation.

6. Close this file and open **Rock_box_full.dem** from the folder.

EDEM Creator: Setting up the model

Step 1: Define the Base Particles

All types of particle used in a model are defined in the Bulk Material section.

In this part you need to create two multi-sphere shapes and one Polyhedral.

Create new particle types

1. Select and expand **rock** from the Bulk Material section.
2. Right click on the GEMM particle shape and select Delete Particle. Click Ok to the warning message about deleting the associated factory.
3. Right click **rock** and select Add Particle > Add Multi-Sphere and then enter the name **rock particle_1** into the highlighted field, repeat to add another Multi-Sphere particle **rock particle_2**
4. Create a 3rd shape this time to Add Particle > Add Polyhedral named **rock particle_3**

Define the spheres and properties

Multi-Sphere Particles are made up of one or more spheres. In this case we import CAD templates of rock particles of different shapes and fit spheres to the template.

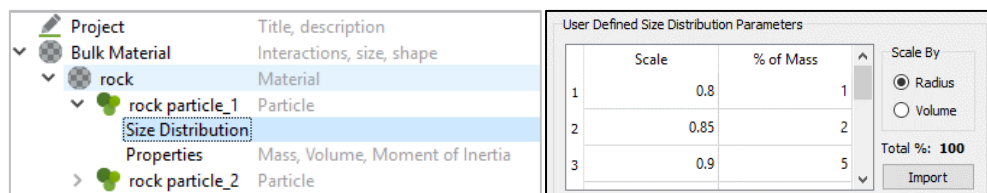
1. Select Tools > Options... > Rendering Options > Display Templates.
2. Click the Import button. You need to import three particle templates **Rock_Template1.stl** and **Rock_Template 2.stl**
3. When Import Options dialog appears, set the measurement units to **Millimeters**, leave all the settings at the default values and click OK.
4. Click the  button and rename the templates to **Rock_Template1** and **Rock_Template2** then click OK.

5. To view the template, select **rock particle_1** from the Creator Tree and on the lower right corner, from the Display Templates drop-down list, select **Rock_Template1**, note that the template will appear large compared to the spheres until step 6 is completed.
6. Select **rock_particle1** within the Bulk Material section.

Modify the rock particle_1 Spheres section at the bottom of the screen. The radii and positions of spheres are presented in the table below:

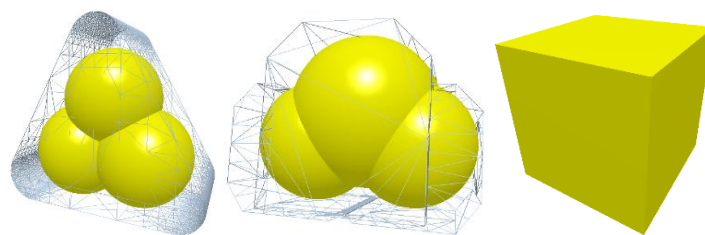
Particle Name	rock particle_1			rock particle_2		
Sphere	0	1	2	0	1	2
Radius (mm)	25	25	25	25	20	20
Position X (mm)	0	-17	17	0	16	-16
Position Y (mm)	19	-9	-9	2	-8	-8
Position Z (mm)	0	0	0	0	0	0

7. Set the Size Distribution to **user defined**, click the Import button and choose PSD.csv. This will set the size distribution as set out in the csv file.



8. Select the **Properties** section and click the **Calculate Properties** button.
9. Repeat instructions 6, 7, 8 for the second multi-sphere particle
10. For rock particle_3 apply the same user defined Size Distribution then under properties select **Auto Calculation**.
11. Set the Cuboid Box Length X,Y,Z values to 50 mm

The shapes that you should obtain along with the templates are presented below. Note that we have created three different particles to highlight the variety of rock particles shapes.



Step 2: Create the Particle Factory

Create the particle factory

1. Expand the Geometries section, right click on the **factory_plate** and select **Add Factory**.

Set the factory's particle creation values

In this model, all three rock particles are to be created at a rate of 100 kg per second. Therefore:

1. Check that the Factory Type is set to **dynamic** and select **Unlimited Number**.
2. In the Generation Rate section, select **Target Mass** and set the rate to **300 kg/s**.

To ensure that this corresponds to each particle being generated at 100 kg/s, select **rock** in the Bulk Material section of the Creator Tree and check that the Particle Ratios section shows an even composition of the three rock particle types (33.33333% each).

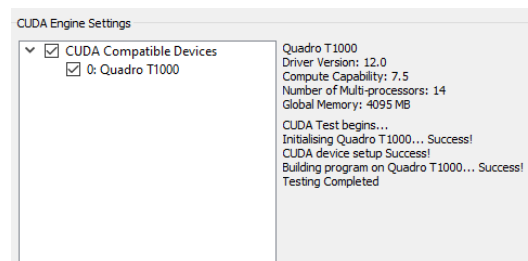
Set the factory's initial parameters

1. Select the newly created factory (e.g. New Factory 2) in the Creator Tree.
2. Check that the Type is set to **dynamic**. If the type is set to Static, right click the factory and select Change Factory Type.
3. Set the Velocity to **fixed** then click the configure button  to set the velocity in the X-direction to 3.54m/s.

EDEM Simulator: Running Simulation

Once the model has been set-up in the Creator we move to the Simulator to run it. Note this simulation requires a Nvidia GPU card. To setup the CUDA Engine:

1. Go to Tools > Options > Simulator Engine > CUDA
2. Select your Nvidia GPU if shown, otherwise select Device Start Test and look for the Success! Test results.



3. Select OK and move to the Simulator 

Step 1: Set the Time Options

Set the time step

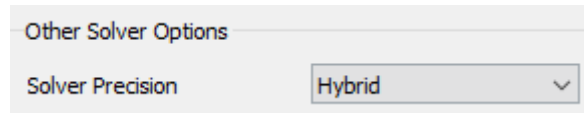
1. If checked, **uncheck** the Auto Time Step checkbox.
2. Set the Time Step value to 0.0001 s, this should be approximately 12% of the Rayleigh Time Step

Set the simulation time and data write-out frequency

The simulation time is the amount of real time your simulation represents.

1. In Simulation Time section, set the Total Time to **15s**.

2. Set the Target Save Interval to **0.02s** to specify write-out frequency.
3. As this is a mixed shape solver with Polyhedral particles the GPU engine and settings are disabled. This simulation is to be run on CUDA GPU only.
4. Under Advanced Settings choose solver precision Hybrid.
 - a. Double precision is a direct match to the CPU solver which only has the double precision option. Many graphics cards have slower double precision calculation speed than single precision, the mixed shape solver allows a mix of single and double precision, only using double when required for accuracy.



Step 3: Run the Simulation

You are now ready to start the simulation.

1. Click the start Progress button  at the bottom of the Viewer window.
2. In Viewer Controls, click the **Refresh Viewer** button  or enable **Auto Update**  at any point to see how the simulation is progressing.
3. This simulation takes 15-30 minutes to run depending on the GPU card.

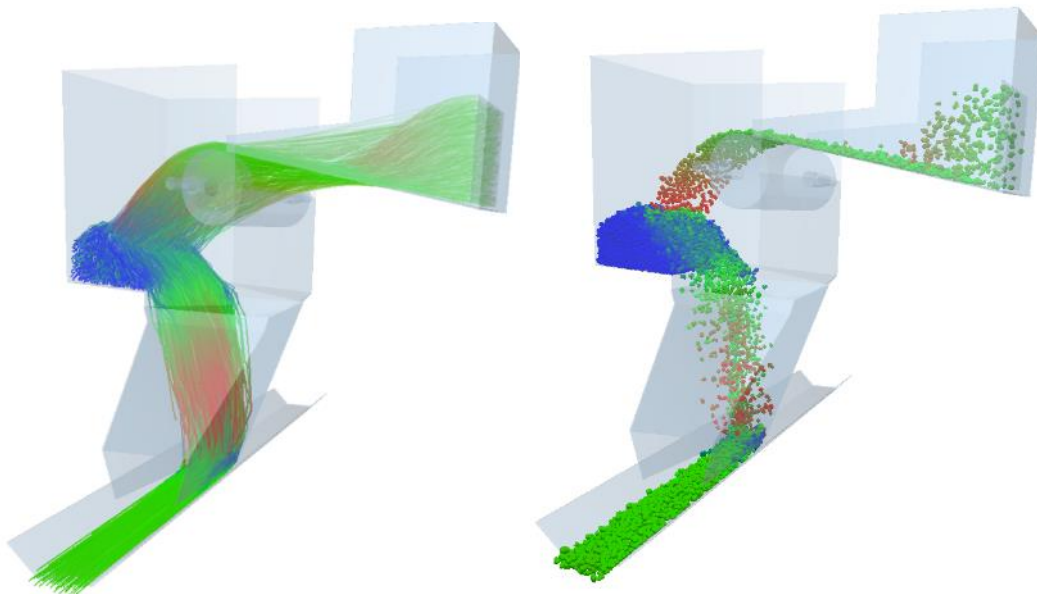
EDEM Analyst: Analyzing Your Results

The Analyst is used to review, examine and analyze the results of the simulation. Part of the analysis has been performed in Part 1 already. Refer to instructions starting at page 14.

Step 1: Configuring the Display

Configure the particles

1. In the Particles section of the Analyst Tree, go to Representation pull-down menu and experiment with changing it to **Vector**, **Cone** or **Streams** as before.
2. Click the **Apply All** button.
3. Use the controls that are located at the bottom of the viewer window to watch the simulation. The images below show Stream and Default particle representations.



4. Expand the Particles section in the Analyst Tree.
 5. Select **rock particle_1**.
 6. Set Representation to Template.
 7. Click Options...
 8. Set the Template to **Rock_Template 1**.
 9. If present, click Apply.
 10. Repeat for **rock particle_2** and set associated template **Rock_Template 2**
- The particles are now displayed by their associated CAD templates.

Step 2: Coloring Elements

Color the particles according to their velocity

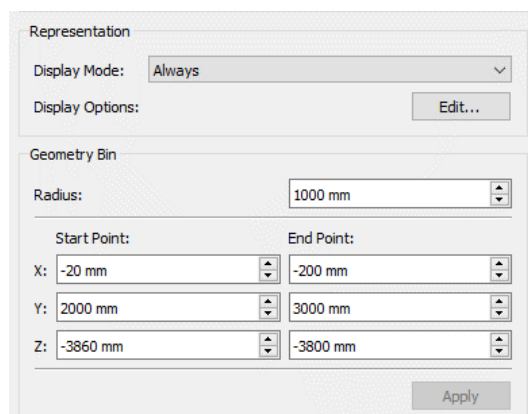
1. Select **rock particle_1**.
2. In the Coloring section, set the Color by to **Velocity**.
3. Set the Min, Mid and Max colors to the **Blue, Green** and **Red** respectively.
4. Click the buttons  next to the Min and Max Value fields – this reads the current values from the point in the simulation being displayed in the Viewer at the moment. Unselect Auto Update and set minimum to **0m/s** and maximum to **7m/s**.
5. If present, click the Apply button to color the particles. If the Auto Update Display feature is enabled, the display will automatically reflect these changes and no Apply button will appear.
6. Check the **Show Legend** box then click Apply.
7. Repeat steps 1 to 6 for **rock particle_2** and **rock_particle_3**.
8. Rewind the simulation, if necessary, and click the Play button to view the distribution of velocity over time.

Step 3: Modifying Selections

The instructions of how to create the mass flow sensor and grid bin group can be found in Part 1 (**page 16**). If necessary, recreate these two selections. The queries should stay the same as previously created.

Modify the mass flow sensor

1. Select **Mass Flow Sensor 01** from the Analyst Tree.
2. Modify the properties of the bin group as follows then click Apply:



Representation	
Display Mode:	Always
Display Options:	<button>Edit...</button>
Geometry Bin	
Radius:	1000 mm
Start Point:	
X:	-20 mm
Y:	2000 mm
Z:	-3860 mm
End Point:	
X:	-200 mm
Y:	3000 mm
Z:	-3800 mm
<button>Apply</button>	

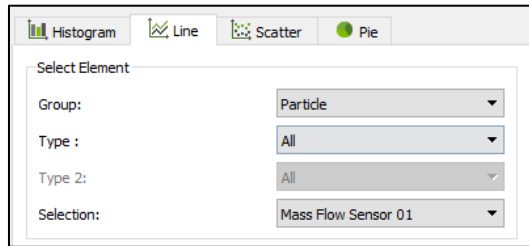
Step 4: Plotting the graphs

New graphs need to be created for this configuration.

Plot a graph of mass flow rate on the bottom conveyor

To plot a linear graph of mass flow rate of all particles changing over time on the bottom conveyor:

1. Click the Analyst button on the toolbar, then click the Create Graph button .
2. Select the Line Graph tab, set the parameters as follows:



Select Element

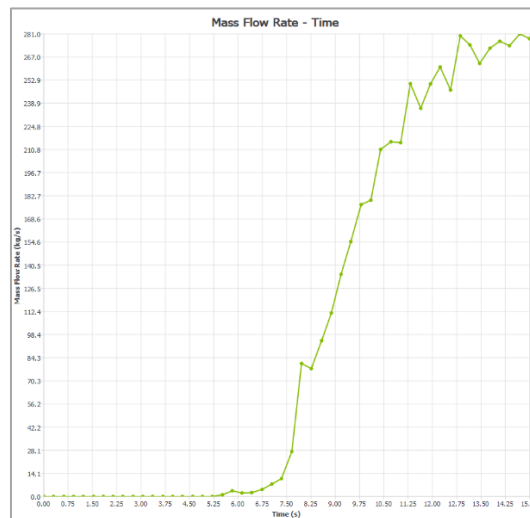
Group: Particle

Type: All

Type 2: All

Selection: Mass Flow Sensor 01

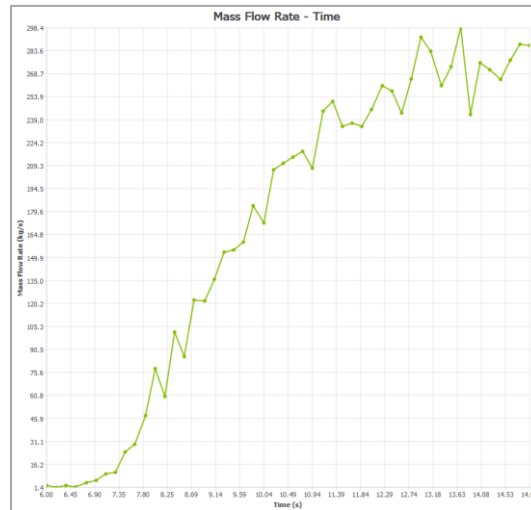
3. In the X-axis tab, ensure that the Time Range Start and End fields cover the whole simulation time.
4. Click the Y-axis tab. Set the Primary Attribute to **Mass Flow Rate**.
5. Click Create Graph at the bottom to plot the graph.



It can be observed from the graph that the first particles flow through the sensor after about 5.5 seconds of simulation, and that a steady-state may not yet have been reached after 10 seconds.

To get a more detailed view of the end of the plot:

6. In the X-axis tab, set the Start time to about **6s**. Click Create Graph to see the new plot.
7. In order to generate the graph of mass flow rate when the system reaches steady state the simulation can be run for longer.



Plot the graph of total particle mass in the rock box

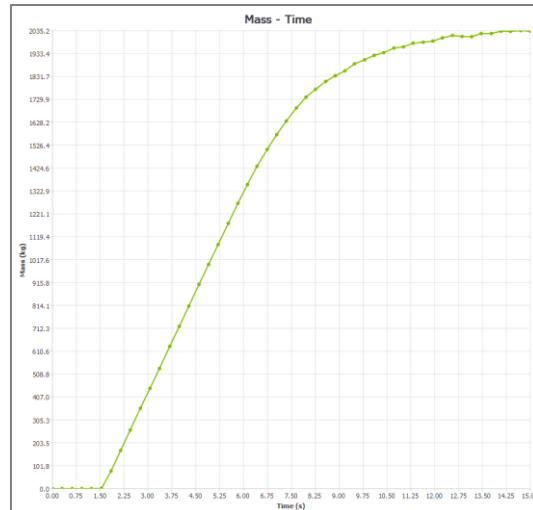
The bin group over the rock box registers the change of total mass of the particles there. To plot the increase in mass over time in the rock box:

1. In the Line Graph tab, set the parameters as follows:

Line Graph Tab Settings:

- Group: Particle
- Type: All
- Type 2: All
- Selection: Rock_Box_Group

2. In the X-axis tab, ensure that the Time Range Start and End fields cover the whole simulation time
3. Click the Y-axis tab. Set the Primary Attribute to **Mass** and the Component to **Total**.
4. Click Create Graph at the bottom to plot the graph.

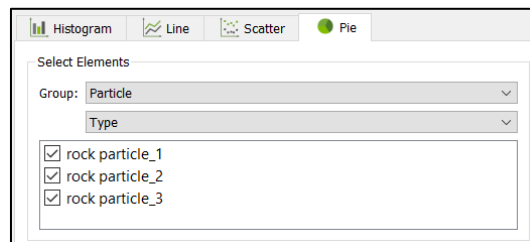


- Experiment with changing the Start time in the X-tab to different values.

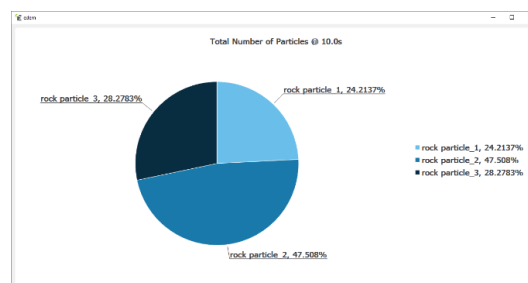
Plot a pie chart of the percentage of each particle type in the rock box

The **Rock_Box Group** can also be used to analyze the content of different particle types in the rock box. To create a pie chart of the distribution:

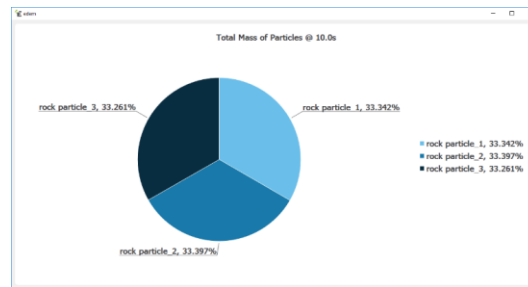
- Choose the Pie Chart tab and set the parameters as follows:



- In the Define Attribute tab, set the Attribute to **Number of Particles** and select **Total** as the Component.
- In the Settings tab, leave the checkbox **Current Time Step** checked. The number of particles in the time step in which you stopped the simulation will be shown.
- Click the **Create Graph** button. The image below was taken at the final timestep of 10.0s. It can be seen that the most common type of particle is rock particle_2.



- Now change the Attribute to **Mass**, leaving the Component set to **Total** as done previously.
- Click the Create Graph button to see a new pie chart for the same time step.



Notice that the distribution of mass is approximately equal for all particles types, as expected. The previous graph showed that the most common particles was rock particle_2, as these are the least dense of the three particle types; therefore, more particles are needed to match the specified mass flow rate (100kg/s per particle type) than for the other two particle types.